第五章 基本应用二

§5.1 中心力场

1. 一般性质

旋转对称性，角动量守恒：$[L, \hat{H}] = 0$

$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(r)$

$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$

定态 Schrodinger 方程：

$$-\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}\right] + V\psi = E\psi$$

分离变量：$\psi(r,\theta,\phi) = R(r)Y(\theta,\phi)$

$$\left\{\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)\frac{1}{R} - \frac{2m}{\hbar^2}[V(r) - E]\right\} + \left[\frac{1}{Y\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) + \frac{1}{Y\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2}\right]\frac{1}{Y} = 0$$

角度部分：$\frac{\partial^2 Y}{\partial\theta^2} + \sin\theta\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) - c\sin^2\theta Y = 0$

其中 $c = l(l+1)$ 解为 $Y_{lm}(\theta,\phi)$

径向部分：$\frac{\partial}{\partial r}\left(r^2\frac{\partial R_l}{\partial r}\right) - \frac{2mr^2}{\hbar^2}\left[V(r) + \frac{l(l+1)\hbar^2}{2mr^2} - E\right]R_l = 0$

令 $u_l = r\cdot R_l$

$$\frac{\partial}{\partial r}\left[r^2\frac{\partial}{\partial r}\left(\frac{u_l}{r}\right)\right] = \frac{\partial}{\partial r}r^2\frac{1}{r}\left[\frac{\partial u_l}{\partial r}\cdot r - u_l\right] = \frac{\partial^2 u_l}{\partial r^2}r + \frac{\partial u_l}{\partial r} - \frac{\partial u_l}{\partial r} = r\frac{\partial^2 u_l}{\partial r^2}$$

$$\Rightarrow -\frac{\hbar^2}{2m}\frac{\partial^2 u_l}{\partial r^2} + \left[V(r) + \frac{l(l+1)\hbar^2}{2mr^2}\right]u_l = E u_l$$

束缚能量与 m 无关，是 $2l+1$ 重简并的。

2. 角动量

$[\hat{H}, \hat{L}^2] = 0$, $[\hat{H}, \hat{L}_z] = 0$ ($[\hat{L}^2, \hat{L}_z] = 0$)

考虑本征态为 $|E, l, m\rangle$

$\langle \vec{r} | E, l, m\rangle = R_{El}(r) Y_{lm}(\theta, \phi)$

满足球 Bessel 方程：

$$\frac{d^2 R_{El}(r)}{dr^2} + \frac{2}{r}\frac{dR_{El}}{dr} + \left[\frac{2mE}{\hbar^2} - \frac{l(l+1)}{r^2}\right] R_{El} = 0$$

$\frac{2mE}{\hbar^2} = k^2$

通解为 $R_{El}(r) = A j_l(kr) + B n_l(kr)$

$$= A(-1)^l x^l \left(\frac{1}{x}\frac{d}{dx}\right)^l \frac{\sin x}{x} + B(-1)^{l+1} x^l \left(\frac{1}{x}\frac{d}{dx}\right)^l \frac{\cos x}{x}$$

要求 $\lim_{r\to 0} R_{El}(r) = 0$, $\lim_{r\to 0} j_l(kr) = \infty$, 则 $R_{El}(r) = A j_l(kr)$

$\langle \vec{r} | E, l, m\rangle = A j_l(kr) Y_{lm}(\theta, \phi)$

归一化：$\langle E, l, m | E', l', m'\rangle = \delta(E - E')\delta_{ll'}\delta_{mm'}$

$$= \int_V \psi_{E,l,m}\, \psi_{E',l',m'}\, r^2 \sin\theta\, dr\, d\theta\, d\phi$$

$$= \int_0^\infty R_{E,l}(r) R_{E',l'}(r)\, r^2 dr \int_0^{2\pi}\int_0^{\pi} Y_{lm}(\theta,\phi) Y_{l'm'}(\theta,\phi) \sin\theta\, d\theta\, d\phi$$

$$= \int_0^\infty R_{E,l}(r) R_{E',l'}(r)\, r^2 dr \cdot \delta_{ll'}\delta_{mm'}$$

$\Rightarrow \int_0^\infty R_{E,l}(r) R_{E',l'}(r)\, r^2 dr = \delta(E - E')$ $\quad k = \frac{\sqrt{2mE}}{\hbar}$

$\frac{1}{k^2}\int_0^\infty j_l(kr) j_{l'}(k'r)\, r^2 dr = \left(\frac{dE}{dk}\right)^{-1}\delta(k - k') = \frac{\hbar}{mk\hbar^2}\delta(k - k')$

$$\int_0^\infty j_l(kr) j_{l'}(k'r)\, r^2 dr = \frac{\pi}{2k^2}\delta(k - k')$$

$\Rightarrow A = \sqrt{\frac{2k^2}{\pi}}$ ($m = \hbar = 1$) ?

$r \to \infty$：$j_l(kr)$ 渐进形式

$$j_l(kr) \xrightarrow{r\to\infty} \frac{1}{kr}\sin\left(kr - \frac{l\pi}{2}\right)$$

$$= \frac{1}{2ikr}\left(e^{-i\frac{l\pi}{2}}\frac{e^{ikr}}{r} - e^{i\frac{l\pi}{2}}\frac{e^{-ikr}}{r}\right)$$

发散球面波　　会聚球面波

* 平面波 ↔ 球面波

将平面波在球面波基矢下展开

$$|k\rangle = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} C_{lm} |E, l, m\rangle$$

$$\langle r|k\rangle = \sum_{l,m} C_{lm} \langle r|E, l, m\rangle$$

$$\frac{e^{ikr\cos\theta}}{(2\pi)^{3/2}} = \sum_{l,m} C_{lm} j_l(kr) Y_{lm}(\theta, \phi)$$

$\vec{k}$ 沿 z 轴 → 与 m 无关，只保留 $m=0$ 项

$$\frac{e^{ikr\cos\theta}}{(2\pi)^{3/2}} = \sum_{l} C_{l0} j_l(kr) Y_{l0}(\theta, \phi)$$

3. 原子轨道

设 r_1, r_2 分别为原子核、电子的坐标

$$\hat{H} = -\frac{\hbar^2}{2m_1}\nabla_1^2 - \frac{\hbar^2}{2m_2}\nabla_2^2 + V(|\vec{r}_1 - \vec{r}_2|)$$

利用质心运动定律，可转化为单体问题

质心坐标 $\vec{R} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1+m_2}$，相对坐标 $\vec{r} = \vec{r}_1 - \vec{r}_2$

总质量 $m = m_1 + m_2$，约化质量 $\mu = \frac{m_1 m_2}{m_1+m_2}$

以下记 $\nabla_r = \nabla$

$$\nabla_1 = \nabla + \frac{m_1}{m_1+m_2}\nabla_R \qquad r_1 = r_1(R, r), \ \frac{\partial}{\partial r_1} = \frac{\partial}{\partial R}\frac{\partial R}{\partial r_1} + \frac{\partial}{\partial r}\frac{\partial r}{\partial r_1}$$

$$\nabla_1^2 = \nabla^2 + \left(\frac{m_1}{m_1+m_2}\right)^2 \nabla_R^2 + \frac{m_1}{m_1+m_2}(\nabla\cdot\nabla_R + \nabla_R\cdot\nabla)$$

$$\nabla_2 = -\nabla + \frac{m_2}{m_1+m_2}\nabla_R$$

$$\nabla_2^2 = \nabla^2 + \left(\frac{m_2}{m_1+m_2}\right)^2 \nabla_R^2 - \frac{m_2}{m_1+m_2}(\nabla\cdot\nabla_R + \nabla_R\cdot\nabla)$$

则 $\hat{H} = -\frac{\hbar^2}{2}\left(\frac{1}{\mu}\nabla^2 + \frac{1}{m}\nabla_R^2\right) + V(r)$

$$= -\frac{\hbar^2}{2m}\nabla_R^2 - \frac{\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0}\frac{1}{r}$$

分离变量：$\Psi(R, r) = \phi(R)\psi(r)$

$$\begin{cases} -\frac{\hbar^2}{2m}\nabla_R^2\phi(R) = E_2\phi(R) \\ -\frac{\hbar^2}{2\mu}\nabla^2\psi(r) - \frac{Ze^2}{4\pi\varepsilon_0}\frac{1}{r}\psi(r) = (E_{tot} - E_2)\psi(r) = E\psi(r) \end{cases}$$

对于相对运动部分，束缚态可表为

$$\psi(\vec{r}) = R_{nl}(r)\,Y_{lm}(\theta,\phi)$$

径向：$-\frac{\hbar^2}{2\mu r^2}\frac{d}{dr}\left[r^2\frac{dR_{nl}(r)}{dr}\right] + \left[-\frac{ze^2}{4\pi\varepsilon_0}\frac{1}{r} + \frac{\hbar^2}{2\mu}\frac{l(l+1)}{r^2}\right]R_{nl} = E_n R_{nl}$

代数解法：

令 $\hat{H}_r R_{nl}(r) = E R_{nl}(r)$　　$a = \frac{4\pi\varepsilon_0\hbar^2}{\mu e^2}$

其中 $\hat{H}_r = \frac{\hat{P}_r^2}{2\mu} + U_{eff}(\hat{r})$，$\hat{P}_r = -i\hbar\left(\frac{\partial}{\partial r} + \frac{1}{r}\right)$，$U_{eff}(r) = -\frac{ze^2}{4\pi\varepsilon_0}\frac{1}{r} + \frac{\hbar^2}{2\mu}\frac{l(l+1)}{r^2}$

推导：

$$-\frac{1}{2\mu}\frac{\hbar^2}{r^2}\frac{d}{dr}\left[r^2\frac{dR_{nl}(r)}{dr}\right] + \left[-\frac{ze^2}{4\pi\varepsilon_0}\frac{1}{r} + \frac{\hbar^2}{2\mu}\frac{l(l+1)}{r^2}\right]R_{nl} = E R_{nl}$$

$$= -\frac{\hbar^2}{2\mu}\left[\frac{d^2R_{nl}(r)}{dr^2} + \frac{2}{r}\frac{dR_{nl}(r)}{dr}\right] + \left[-\frac{ze^2}{4\pi\varepsilon_0}\frac{1}{r} + \frac{\hbar^2}{2\mu}\frac{l(l+1)}{r^2}\right]R_{nl} = E R_{nl}$$

$$= \frac{(-i\hbar)^2}{2\mu}\left[\frac{d^2R_{nl}(r)}{dr^2} + \frac{1}{r}\frac{dR_{nl}(r)}{dr} + \frac{d}{dr}\left[\frac{1}{r}R_{nl}(r)\right] + \frac{1}{r^2}R_{nl}\right] + \underbrace{\left[-\frac{ze^2}{4\pi\varepsilon_0}\frac{1}{r} + \frac{\hbar^2}{2\mu}\frac{l(l+1)}{r^2}\right]}_{U_{eff}}R_{nl} = E R_{nl}$$

$$= \left[\frac{1}{2\mu}\underbrace{(-i\hbar)^2\left(\frac{d}{dr} + \frac{1}{r}\right)^2}_{\hat{P}_r^2} + U_{eff}(\vec{r})\right]R_{nl}(r) = E R_{nl}(r)$$

证明：

(1) $\left[\hat{r}, \hat{P}_r\right] = i\hbar$

证：$\left[\hat{r}, \hat{P}_r\right]f(r) = -i\hbar\left[r\cdot\frac{d}{dr} + \frac{1}{r}\right]f(r)$

$$= -i\hbar\left\{r\frac{df}{dr} + f(r) - \frac{d}{dr}\left[rf(r)\right] - f(r)\right\}$$

$$= -i\hbar\left[-f(r)\right] = i\hbar f(r)$$

(2) $\left[\hat{P}_r, \frac{1}{r}\right] = \frac{i\hbar}{r^2}$

证：$\left[\hat{P}_r, \frac{1}{r}\right)f(r) = -i\hbar\left[\frac{d}{dr} + \frac{1}{r}, \frac{1}{r}\right]f(r)$

$$= -i\hbar\left\{\frac{d}{dr}\left(\frac{f(r)}{r}\right) + \frac{f(r)}{r^2} - \frac{1}{r}\frac{df(r)}{dr} - \frac{f(r)}{r^2}\right\}$$

$$= -i\hbar\left\{\frac{\frac{df(r)}{dr}\cdot r - f(r)}{r^2} + \frac{f(r)}{r^2} - \frac{1}{r}\frac{df(r)}{dr} - \frac{f(r)}{r^2}\right\}$$

$$= \frac{i\hbar}{r^2}f(r)$$

定义算符 $\hat{a}_l$, $\hat{a}_l^+$

$$\begin{cases} \hat{a}_l = \frac{\mu\omega}{2\hbar}\left[\hat{r} - \frac{\hbar(l+1)}{\mu\hat{r}}\right] + \frac{\hat{p}_r}{\mu\omega} \\ \hat{a}_l^+ = \frac{\mu\omega}{2\hbar}\left[\hat{r} - \frac{\hbar(l+1)}{\mu\hat{r}}\right] - \frac{\hat{p}_r}{\mu\omega} \end{cases}$$

(1) $[\hat{a}_l, \hat{a}_l^+] = \frac{1}{2\hbar}[\hat{r}\hat{p}_r - \hat{p}_r\hat{r}] = \frac{1}{2\hbar}[\hat{r}\hat{p}_r - \hat{p}_r\hat{r}] = 1$

(2) $[\hat{a}_l^+, \hat{a}_l] = -1$

$$= \frac{\mu\omega}{2\hbar}\left[\hat{r} - \frac{\hbar(l+1)}{\mu\hat{r}}, \hat{r} - \frac{\hbar(l+1)}{\mu\hat{r}}\right] = \frac{\mu\omega}{2\hbar}\left[\frac{\hbar(l+1)}{\mu\hat{r}}, \hat{r}\right] = \frac{\hbar}{2}$$

$$= \frac{\hbar}{2}\left[\frac{\hbar(l+1)}{\mu\hat{r}}, \hat{r}\right] = \frac{\hbar}{2}\left[\frac{\hbar(l+1)}{\mu}\cdot\frac{1}{\hat{r}}, \hat{r}\right] = \frac{\hbar}{2}\cdot\frac{\hbar(l+1)}{\mu}\left[\frac{1}{\hat{r}}, \hat{r}\right] = \frac{\hbar}{2}\cdot\frac{\hbar(l+1)}{\mu}\cdot\frac{1}{\hat{r}}[\hat{r}, \hat{r}] = 0$$

(3) $\hat{H}_{l+1} - \hat{H}_l = \frac{\hbar\omega}{2}\left\{\left[\frac{(l+1)^2}{(l+1)^2} - \frac{(l+1)^2}{(l+2)^2}\right](l+1)\right\} = \frac{\hbar\omega}{2}(l+1)$

$$\hat{H}_{l+1} = \frac{\hbar\omega}{2}\left[\hat{a}_{l+1}\hat{a}_{l+1}^+ - \frac{(l+1)^2}{(l+2)^2}\right]$$

$$\hat{a}_{l+1}\hat{a}_{l+1}^+ = \frac{\mu\omega}{2\hbar}\left[\hat{r} - \frac{\hbar(l+2)}{\mu\hat{r}}\right]\cdot\frac{\mu\omega}{2\hbar}\left[\hat{r} - \frac{\hbar(l+2)}{\mu\hat{r}}\right] = \frac{\mu^2\omega^2}{4\hbar^2}\left\{\hat{r}^2 + \left[\frac{\hbar(l+2)}{\mu\hat{r}}\right]^2 - \frac{\hbar(l+2)}{\mu}\left[\frac{1}{\hat{r}}\hat{r} + \hat{r}\frac{1}{\hat{r}}\right]\right\}$$

$$= \frac{\mu^2\omega^2}{4\hbar^2}\left\{\frac{\hbar^2}{\mu^2\omega^2}\hat{p}_r^2 + \frac{\hbar^2(l+2)^2}{\mu^2\hat{r}^2} - \frac{\hbar(l+2)}{\mu}\left[\frac{i\hbar}{\mu\hat{r}^2}\hat{r} + \hat{r}\frac{i\hbar}{\mu\hat{r}^2}\right]\right\}$$

$$\frac{2\mu}{\hbar^2}\left[\hat{a}_l^+\hat{a}_l - \frac{(l+1)^2}{2}\right] = \frac{2\mu}{\hbar^2}\cdot\frac{\hbar^2}{2\mu}\left[\frac{1}{\hat{r}^2} - \frac{\hbar(l+1)}{\mu\hat{r}^2}\right] = \frac{1}{\hat{r}^2} - \frac{\hbar(l+1)}{\mu\hat{r}^2}\left[\frac{i\hbar}{\mu\hat{r}^2}\hat{r} + \hat{r}\frac{i\hbar}{\mu\hat{r}^2}\right]$$

$$= \frac{\hat{p}_r^2}{2\mu} + \frac{1}{2\mu}\left[\frac{\hbar^2(l+1)^2}{\mu\hat{r}^2} - \frac{i\hbar}{\mu\hat{r}^2}\hat{r}\right]$$

$$= \frac{\hat{p}_r^2}{2\mu} + \frac{\hbar^2(l+1)^2}{2\mu\hat{r}^2} - \frac{i\hbar}{2\mu\hat{r}^2}\hat{r} = \hat{H}_l$$

$-i\hbar(l+1)^2$

(4) $[\hat{a}_l, \hat{H}_l'] = [\hat{H}_{l+1}' - \hat{H}_l']\hat{a}_l$

证：$[\hat{a}_l, \hat{H}_l'] = \frac{\hbar^2}{\mu a^2}[\hat{a}_l, \hat{a}_l^+\hat{a}_l - \frac{l^2}{2}]$

$$= \frac{\hbar^2}{\mu a^2}[\hat{a}_l, \hat{a}_l^+]\hat{a}_l = \frac{\hbar^2}{\mu a^2}\cdot\frac{\mu a^2}{\hbar^2}(\hat{H}_{l+1}' - \hat{H}_l')\hat{a}_l$$

$$= (\hat{H}_{l+1}' - \hat{H}_l')\hat{a}_l$$

(5) $\hat{a}_l\hat{H}_l' = \hat{H}_{l+1}'\hat{a}_l$

由(4)易得.

$R_{Z,l+1} = \hat{C}_l^+ R_{Z,l}$ 也是能量 $\hat{H}$ 本征态，本征值为 E

l 存在一上限 l_{max}

$$\hat{C}_{l_{max}} R_{Z,l_{max}} = 0$$

$$\hat{H}_{l_{max}} R_{Z,l_{max}} = \frac{\hbar^2}{\mu a^2}\left[\hat{C}^+_{l_{max}} \hat{C}_{l_{max}} - \frac{Z^2}{2(l+1)^2}\right] R_{Z,l_{max}}$$

$$= -\frac{Z^2\hbar^2}{2\mu a^2 (l_{max}+1)^2} R_{E,l_{max}} = E R_{Z,l_{max}}$$

记 $n = l_{max}+1$

$$E_n = -\frac{Z^2\hbar^2}{2\mu a^2 n^2} = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{2a}\frac{1}{n^2}$$

近似认为 $\mu \approx m_e$

$$a = a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}, \text{ Bohr 半径}$$

$n = 1, 2, 3, \cdots$

给定 n，$l = 0, 1, \cdots n-1$

对于能级 E_n，简并度为 $d = \sum_{l=0}^{n-1}(2l+1) = n^2$

求波函数的具体形式：

$$\psi_{nlm}(r,\theta,\varphi) = R_{nl}(r) Y_{lm}(\theta,\varphi)$$

给定 n，$\hat{C}_{n-1} R_{n,n-1}(r) = 0$

$$\frac{a}{r}\left(\frac{\partial}{\partial r} - \frac{n}{r} + \frac{Z}{na}\right) R_{n,n-1}(r) = 0$$

解得 $R_{n,n-1}(r) = C_{n,n-1} r^{n-1} e^{-\frac{Z}{na}r}$

$\hat{C}_l R_{n,l} = R_{n,l+1}$

$\hat{C}_l^+ R_{n,l+1}(r) = \frac{Z^2}{2}\left[\frac{1}{(l+1)^2} - \frac{1}{n^2}\right] R_{n,l}(r)$ ← $\hat{C}_l^+ \hat{C}_l R_{n,l} = \hat{C}_l R_{n,l+1}$

$R_{n,l-1}(r) = \frac{2n^2 l^2}{Z^2(n+l)(n-l)} \hat{C}_{l-1} R_{n,l}(r)$ $\quad \hat{C}_l R_{n,l+1} = \cdots ?$

$$\Rightarrow R_{n,l}(r) = \sqrt{\left(\frac{2Z}{na}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3}}\, e^{-\frac{\rho}{2}} \rho^l L^{2l+1}_{n-l-1}(\rho)$$

§3.2 角动量.

1. 角动量表示

$\vec{S} = (\hat{S}_x, \hat{S}_y, \hat{S}_z)$ 选取 $(\hat{S}^2, \hat{S}_z)$ 表象.

$S = \frac{1}{2}$, $S_z = \frac{1}{2}\hbar$ / $-\frac{1}{2}\hbar$

表示为 $|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\hat{S}_z = \frac{1}{2}\hbar \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ $\hat{S}_z|\uparrow\rangle = \frac{1}{2}\hbar|\uparrow\rangle$, $\hat{S}_z|\downarrow\rangle = -\frac{1}{2}\hbar|\downarrow\rangle$

定义梯子算符

$\hat{S}_\pm = \hat{S}_x \pm i\hat{S}_y$

$\hat{S}_+|\uparrow\rangle = 0$, $\hat{S}_+|\downarrow\rangle = \hbar|\uparrow\rangle$

$\hat{S}_-|\uparrow\rangle = \hbar|\downarrow\rangle$, $\hat{S}_-|\downarrow\rangle = 0$

$\Rightarrow$ $\hat{S}_+ = \hbar\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\hat{S}_- = \hbar\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$

$\hat{S}_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\hat{S}_y = \frac{\hbar}{2i}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

Pauli 矩阵:

$\vec{S} = (\hat{S}_x, \hat{S}_y, \hat{S}_z)$

$= \frac{1}{2}\hbar\vec{\sigma} = \frac{1}{2}\hbar(\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z)$

$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ $\hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

对易关系: $[\hat{\sigma}_\alpha, \hat{\sigma}_\beta] = 2i\varepsilon_{\alpha\beta\gamma}\hat{\sigma}_\gamma$

反对易关系: $\{\hat{\sigma}_\alpha, \hat{\sigma}_\beta\} = 2\delta_{\alpha\beta}$

例: 在 $\hat{S}_z$ 表象中, 求 $\vec{S}\cdot\vec{n}$ 的本征态, 其中 $\vec{n}$ 是任意单位矢量.

设 $\vec{n} = \sin\theta\cos\varphi\,\vec{x} + \sin\theta\sin\varphi\,\vec{y} + \cos\theta\,\vec{z}$

$\hat{S}\cdot\hat{n} = \frac{1}{2}\hbar\left|\sin\theta\cos\varphi\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \sin\theta\sin\varphi\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \cos\theta\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right|$

$= \frac{1}{2}\hbar\begin{pmatrix} \cos\theta & e^{-i\varphi}\sin\theta \\ e^{i\varphi}\sin\theta & -\cos\theta \end{pmatrix}$

$\hat{S}\cdot\hat{n}|\psi\rangle = \lambda|\psi\rangle$ $-\frac{\hbar}{2}$ $\frac{\hbar}{2}$

$$\begin{pmatrix} \cos\frac{\theta}{2} & e^{-i\phi}\sin\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2} & -\cos\frac{\theta}{2} \end{pmatrix}$$

$$\begin{vmatrix} \cos\theta-\lambda & e^{-i\phi}\sin\theta \\ e^{i\phi}\sin\theta & -\cos\theta-\lambda \end{vmatrix} = \lambda^2-\cos^2\theta-\sin^2\theta = \lambda^2-1 = 0 \Rightarrow \lambda = \pm 1$$

$\lambda = 1$; $\begin{pmatrix} 1-\cos\theta & e^{-i\phi}\sin\theta \\ e^{i\phi}\sin\theta & -1+\cos\theta \end{pmatrix} \to \begin{pmatrix} 1-\cos\theta & e^{-i\phi}\sin\theta \\ 0 & 0 \end{pmatrix}$

$$\begin{pmatrix} 1-\cos\theta & e^{-i\phi}\sin\theta \\ \frac{(1-\cos\theta)(1+\cos\theta)}{\sin\theta} & \sin\theta \end{pmatrix}$$

$$\textcircled{2} + \textcircled{1}\cdot e^{i\phi}\frac{1-\cos\theta}{\sin\theta}$$

$$|v_1\rangle = \begin{pmatrix} e^{-i\phi}\sin\theta \\ 1-\cos\theta \end{pmatrix} = \begin{pmatrix} e^{-i\phi}\frac{\sin\theta}{1-\cos\theta} \\ 1 \end{pmatrix} = \begin{pmatrix} e^{-i\phi}\frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}} \\ 1 \end{pmatrix} = \begin{pmatrix} e^{-i\phi}\cot\frac{\theta}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} e^{-i\phi}\tan\frac{\pi}{2}-\frac{\theta}{2} \\ 1 \end{pmatrix}$$

$\lambda = -1$; $\begin{pmatrix} 1+\cos\theta & e^{-i\phi}\sin\theta \\ e^{i\phi}\sin\theta & -1-\cos\theta \end{pmatrix} \to \begin{pmatrix} 1+\cos\theta & e^{-i\phi}\sin\theta \\ 0 & 0 \end{pmatrix}$

$$\begin{pmatrix} 1+\cos\theta & e^{-i\phi}\sin\theta \\ \frac{(1+\cos\theta)(1-\cos\theta)}{\sin\theta} & -1-\cos\theta \end{pmatrix}$$

$$|v_2\rangle = \begin{pmatrix} e^{-i\phi}\sin\theta \\ 1+\cos\theta \end{pmatrix} = \begin{pmatrix} e^{-i\phi}\frac{\sin\theta}{1+\cos\theta} \\ 1 \end{pmatrix} = \begin{pmatrix} e^{-i\phi}\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}} \\ 1 \end{pmatrix} = \begin{pmatrix} e^{-i\phi}\tan\frac{\theta}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} e^{-i\phi}\tan\frac{\theta}{2} \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} \cos\frac{\theta}{2} & e^{-i\phi}\sin\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2} & -\cos\frac{\theta}{2} \end{pmatrix}$$

$$\lambda_1 = \frac{1}{2} \quad |v_1\rangle = \begin{pmatrix} e^{-i\frac{\phi}{2}}\cos\frac{\theta}{2} \\ e^{i\frac{\phi}{2}}\sin\frac{\theta}{2} \end{pmatrix}$$

$$\lambda_2 = -\frac{1}{2} \quad |v_2\rangle = \begin{pmatrix} e^{-i\frac{\phi}{2}}\sin\frac{\theta}{2} \\ -e^{i\frac{\phi}{2}}\cos\frac{\theta}{2} \end{pmatrix}$$

2. Kramers 简并.

角动量 = 角动量.

时间反演： $\hat{\Theta}|S,S_z\rangle = (-1)^{S}|S,-S_z\rangle = e^{i2S\pi}|S,-S_z\rangle$

$= \hat{\Theta}^2|S,S_z\rangle = \hat{\Theta}(e^{i2S\pi}|S,-S_z\rangle)$

$= (-1)^{2S_z}(e)^{-2S_z}|S,S_z\rangle = (-1)^{2S_z}|S,S_z\rangle$

$= (-1)^{2S}|S,S_z\rangle$ $(-1)^{2S} = (-1)^{2S_z}$ 对 S_z 成立.

即 对于半整数角动，$\hat{\Theta}^2 = -1$

$[\hat{\Theta},\hat{H}] = 0 \Rightarrow \hat{H}\hat{\Theta}|\psi\rangle = \hat{\Theta}\hat{H}|\psi\rangle = E\hat{\Theta}|\psi\rangle$

对于半整数角动的系统，由，$\hat{H}$ 有出同本征态，存在 kramers 简并

存在两个能量相同的本征态，互为时间反演态.

证：要证 $|\psi\rangle$ 与 $\hat{\Theta}|\psi\rangle$ 是两个不同的态，只需证明 $|\psi\rangle$，$\hat{\Theta}|\psi\rangle$ 正交，

$\langle\psi|\hat{\Theta}|\psi\rangle = \sum_{ij}\langle\psi|i\rangle\langle i|\hat{u}\hat{k}|j\rangle\langle j|\psi\rangle$

$\hat{\Theta}^2 = -1$，

$\hat{\Theta} = -\hat{\Theta}^T$

$\hat{\Theta}^2 = uu^* = -I$

$u^T u^T = -I$ 两边取转置.

$u^T = -u$

$= \sum_{ij}\psi_i^* u_{ij}\hat{k}\psi_j = \sum_{ij}\psi_i^* u_{ij}\psi_j^* = \sum_{ij}\psi_j^*(-u_{ji})\psi_i^*$

$= \sum_{ij}\psi_j^*(-u_{ji})\hat{k}\psi_i = -\langle\psi|\hat{\Theta}|\psi\rangle = 0$

角动量算符性质

$\hat{\Theta}\hat{S}\hat{\Theta}^{-1} = -\hat{S}$，$\hat{\Theta} = \hat{u}\hat{k}$

在 S_z 表象下，$\hat{S}_x$，$\hat{S}_z$ 为实数矩阵，$\hat{S}_y$ 为虚数矩阵.

$\hat{k}$ 使 $\hat{S}_y$ 变号，但不会影响 $\hat{S}_x$，$\hat{S}_z$

猜 $\hat{u} = e^{-\frac{i\alpha\hat{S}_y}{\hbar}}$，绕 y 轴旋转 α.

$\hat{\Theta} = e^{-\frac{i\alpha\hat{S}_y}{\hbar}}\hat{k}$

$= e^{-i\cdot\frac{\pi}{2}\hat{S}_y}\hat{k}$

$= (\cos\frac{\pi}{2} - i\sin\frac{\pi}{2})\hat{S}_y\hat{k} = -i\hat{S}_y\hat{k}$

Stern - Gerlach 实验：

$$\vec{F} = -\nabla(-\vec{\mu}\cdot\vec{B}) = \vec{\mu}\cdot\nabla\vec{B}$$

例：假设用中子做 Stern - Gerlach 实验，磁场强度为 $B_x = B_y = 0$，$B_z = B_0 + \lambda z$，

中子质量为 m，磁矩为 μ，在磁场中的势能为 $\hat{H}_\mu = -\mu\hat{\sigma}\cdot\vec{B}$

在 $t=0$ 时刻，中子的状态可表为 $|\psi_0\rangle = \psi(\vec{r},0)(a|\uparrow\rangle + b|\downarrow\rangle)$

且 $\langle r\rangle = 0$，$\langle p_x\rangle = p_0$，$\langle p_y\rangle = \langle p_z\rangle = 0$

计算经过时间 t 后，自旋向上、下的中子之平均期待值。

解：$\hat{H} = \hat{H}_p + \hat{H}_\mu = \frac{\hat{p}^2}{2m}\begin{pmatrix}1&0\\0&1\end{pmatrix} - \mu B_0 + \lambda\hat{z}\begin{pmatrix}1&0\\0&1\end{pmatrix}$

$[\hat{H}, \hat{S}_z] = 0$ → 可以单独考虑粒子自旋向上、向下的运动

Ehrenfest 定理：

$$\frac{d}{dt}\langle z_\uparrow\rangle = \frac{1}{m}\langle p_{z,\uparrow}\rangle, \quad \frac{d}{dt}\langle p_{z,\uparrow}\rangle = -\nabla V(r) = \mu\lambda$$

$$\frac{d}{dt}\langle z_\downarrow\rangle = \frac{1}{m}\langle p_{z,\downarrow}\rangle, \quad \frac{d}{dt}\langle p_{z,\downarrow}\rangle = -\nabla V(r) = -\mu\lambda$$

$$\Rightarrow \langle z_\uparrow\rangle = \frac{\mu\lambda t^2}{2m}, \quad \langle z_\downarrow\rangle = -\frac{\mu\lambda t^2}{2m}$$

§3.3 角动量相加

$\vec{J} = \vec{J}_1 + \vec{J}_2$，满足角动量的对易关系：

$[\vec{J}_\alpha, \vec{J}_\beta] = i\varepsilon_{\alpha\beta\gamma}\vec{J}_\gamma$

$H = H_1 \otimes H_2$

总角动 $\vec{S} = \vec{S}_1 + \vec{S}_2$，在 $(\vec{S}^2, S_z)$ 表象，

$\vec{S}^2 |s, s_z\rangle = s(s+1)|s, s_z\rangle$，$\vec{S}_z |s, s_z\rangle = s_z |s, s_z\rangle$

选取共同本征矢直积态 $|s\rangle|s_z\rangle = |s, s_z\rangle$

设为 $|\uparrow\uparrow\rangle$，$|\uparrow\downarrow\rangle$，$|\downarrow\uparrow\rangle$，$|\downarrow\downarrow\rangle$

$\vec{S}_z^2 |\uparrow\uparrow\rangle = \hbar |\uparrow\uparrow\rangle$，$\vec{S}_z^2 |\uparrow\downarrow\rangle = 0 |\uparrow\downarrow\rangle$

$\vec{S}_z^2 |\downarrow\downarrow\rangle = -\hbar |\downarrow\downarrow\rangle$，$\vec{S}_z^2 |\downarrow\uparrow\rangle = 0 |\downarrow\uparrow\rangle$

$\vec{S}^2 |\uparrow\uparrow\rangle = 2\hbar^2 |\uparrow\uparrow\rangle$，$\vec{S}^2 |\downarrow\downarrow\rangle = 2\hbar^2 |\downarrow\downarrow\rangle$

$$\begin{cases} \vec{S}^2 |\uparrow\downarrow\rangle = \hbar^2 |\uparrow\downarrow\rangle + \hbar^2 |\downarrow\uparrow\rangle \\ \vec{S}^2 |\downarrow\uparrow\rangle = \hbar^2 |\downarrow\uparrow\rangle + \hbar^2 |\uparrow\downarrow\rangle \end{cases}$$

$$\vec{S}^2 = (\vec{S}_1 + \vec{S}_2)^2 = \vec{S}_1^2 + \vec{S}_2^2 + 2\vec{S}_1 \cdot \vec{S}_2 = \vec{S}_1^2 + \vec{S}_2^2 + 2(\hat{S}_{1x}\hat{S}_{2x} + \hat{S}_{1y}\hat{S}_{2y} + \hat{S}_{1z}\hat{S}_{2z})$$

$$= \vec{S}_1^2 + \vec{S}_2^2 + \frac{1}{2}(\vec{S}_{1+} + \vec{S}_{1-})(\vec{S}_{2-} + \vec{S}_{2+}) - \frac{1}{2}(\vec{S}_{1+} - \vec{S}_{1-})(\vec{S}_{2-} - \vec{S}_{2+}) + 2\vec{S}_{1z}\vec{S}_{2z}$$

$$= \vec{S}_1^2 + \vec{S}_2^2 + \vec{S}_{1+}\vec{S}_{2-} + \vec{S}_{1-}\vec{S}_{2+} + 2\vec{S}_{1z}\vec{S}_{2z}$$

在 $|\uparrow\downarrow\rangle$、$|\downarrow\uparrow\rangle$ 的子空间中，$\vec{S}^2 = \hbar^2 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

$\lambda_1 = 2$，$|\psi_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\lambda_2 = 0$，$|\psi_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\vec{S}^2$ 和 $\vec{S}_z$ 的共同本征态 $|S, S_z\rangle$ 为以下四组：

(1) $S=0$　$|00\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$　两粒子自旋态，交换反对称

(2) $S=1$　$|11\rangle = |\uparrow\uparrow\rangle$

$|11\rangle = |\uparrow\uparrow\rangle$

$|10\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$　自旋三重态，交换对称

$|00\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$

例：

$|\psi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$　$|\psi_2\rangle = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$　求总波函数自旋单重态、三重态的一

$|\psi\rangle = |\psi_1\rangle \otimes (\cos\theta|\downarrow\rangle + \sin\theta|\uparrow\rangle) = \begin{pmatrix} 0 \\ 0 \\ \cos\theta \\ \sin\theta \end{pmatrix}$

$= \cos\theta|\psi_1\downarrow\rangle + \sin\theta|\psi_1\uparrow\rangle$

单重态：$P_1 = |\langle\psi|0,0\rangle|^2 = \frac{1}{2}|\langle\psi_1\downarrow|\cos\theta|\downarrow\uparrow\rangle - \langle\psi_1\uparrow|\sin\theta|\uparrow\downarrow\rangle|^2$

$= \frac{1}{2}\cos^2\theta$

三重态：$P_0 = |\langle 1,-1|\psi\rangle|^2 + |\langle 1,0|\psi\rangle|^2 + |\langle 1,1|\psi\rangle|^2$

$= 0 + \frac{1}{2}\cos^2\theta + \sin^2\theta = \frac{1}{2}\cos^2\theta + \sin\theta$

2. C-G系数.

$\vec{J} = \vec{J}_1 + \vec{J}_2$

记此基矢为 $|j_1, m_1; j_2, m_2\rangle$

$[\vec{J}^2, \vec{J}_1^2] = 0$, $[\vec{J}^2, \vec{J}_2^2] = 0$

存在 $\vec{J}^2$, $\vec{J}_z$, $\vec{J}_1^2$, $\vec{J}_2^2$ 的共同本征空间. $[\vec{J}_z, \vec{J}_1^2] = 0$, $[\vec{J}_z, \vec{J}_2^2] = 0$

$|j, m; j_1, j_2\rangle$

如何用 $|j_1, m_1; j_2, m_2\rangle$ 表示: 展开 C-G系数

$|j, m; j_1, j_2\rangle = \sum |j_1, m_1; j_2, m_2\rangle \langle j_1, m_1; j_2, m_2 | j, m; j_1, j_2\rangle$

给定 j: $m \in [-j, j]$, 有 $2j+1$ 种取值.

给定 m: $m = m_1 + m_2$ $m_1 \in [-j_1, j_1]$, $m_2 \in [-j_2, j_2]$

(m_1, m_2) 有 $\min(j_1, m+j_2) + \min(j_2, m+j_2) - m + 1$ 种取值.

记: $m_1 = m - m_2$, $m_1 \in \min(j_1, m+j_2) - \max(-j_1, m-j_2) + 1$

① $m - j_2 \geq -j_1$, $\max(-j_1, m-j_2) = m - j_2$

$m_1 \in \min(j_1, m+j_2) + j_2 + m + 1 = \min(j_1, m+j_2) + \min(j_2, m+j_2) - m + 1$

② $m - j_2 < -j_1$, $\max(-j_1, m-j_2) = -j_1$

$m_1 \in \min(j_1, m+j_2) + j_1 + 1 = \min(j_1, m+j_2) + \min(j_2, m+j_2) - m + 1$

j 的取值范围.

$j \in j_1 + j_2$. 设 j 最小值为 j_{min}

$\sum_{j=j_{min}}^{j_1+j_2} (2j+1) = (j_1+j_2+1)(j_1+j_2+1)$ 直积态 Hilbert 空间维数.

则 $j_{min} = |j_1 - j_2|$ $(j_{min} \geq 0)$

计算系数：

$$\langle j_1 m_1 j_2 m_2 | J m \rangle = \langle j_1 m_1 j_2 m_2 | (\hat{J}_1^2 + \hat{J}_2^2) | J m \rangle = \langle j_1 m_1 j_2 m_2 | J m \rangle$$

$$= \sum_{m_1' m_2'} \langle j_1 m_1 j_2 m_2 | J m \rangle \langle j_1 m_1' j_2 m_2' | (\hat{J}_1^2 + \hat{J}_2^2) | j_1 m_1' j_2 m_2' \rangle \langle j_1 m_1' j_2 m_2' | J m \rangle$$

$$= \sum_{m_1' m_2'} \langle j_1 m_1 j_2 m_2 | J m \rangle \langle j_1 m_1' j_2 m_2' | [j_1(j_1+1) + j_2(j_2+1)] | j_1 m_1' j_2 m_2' \rangle \langle j_1 m_1' j_2 m_2' | J m \rangle$$

$$= [j_1(j_1+1) + j_2(j_2+1)] \sum_{m_1' m_2'} \langle j_1 m_1 j_2 m_2 | J m \rangle \langle j_1 m_1' j_2 m_2' | J m \rangle$$

$$= [j_1(j_1+1) + j_2(j_2+1)] \langle j_1 m_1 j_2 m_2 | J m \rangle$$

$$\hat{J}_\pm | j m \rangle = \hat{J}_1 \pm \hat{J}_2 | j_1 m_1 j_2 m_2 \rangle = \sqrt{(j \mp m)(j \pm m + 1)} | j_1 m_1 j_2 m_2 \rangle$$

$$\langle j_1 m_1 j_2 m_2 | \hat{J}_\pm | J m \rangle = \sqrt{(j \mp m)(j \pm m + 1)} \langle j_1 m_1 j_2 m_2 | J m \pm 1 \rangle$$

$$= \sum_{m_1' m_2'} \langle j_1 m_1 j_2 m_2 | (\hat{J}_{1\pm} + \hat{J}_{2\pm}) | j_1 m_1' j_2 m_2' \rangle \langle j_1 m_1' j_2 m_2' | J m \rangle$$

$$= \sum_{m_1' m_2'} \langle j_1 m_1 j_2 m_2 | \hat{J}_{1\pm} | j_1 m_1' j_2 m_2' \rangle \langle j_1 m_1' j_2 m_2' | J m \rangle + \sum_{m_1' m_2'} \langle j_1 m_1 j_2 m_2 | \hat{J}_{2\pm} | j_1 m_1' j_2 m_2' \rangle \langle j_1 m_1' j_2 m_2' | J m \rangle$$

$$= \sum_{m_1' m_2'} \sqrt{(j_1 \mp m_1')(j_1 \pm m_1' + 1)} \langle j_1 m_1 j_2 m_2 | j_1 m_1' \pm 1 j_2 m_2' \rangle \langle j_1 m_1' j_2 m_2' | J m \rangle + \sum_{m_1' m_2'} \sqrt{(j_2 \mp m_2')(j_2 \pm m_2' + 1)} \langle j_1 m_1 j_2 m_2 | j_1 m_1' j_2 m_2' \pm 1 \rangle \langle j_1 m_1' j_2 m_2' | J m \rangle$$

$$= \sum_{m_1'} \sqrt{(j_1 \mp m_1')(j_1 \pm m_1' + 1)} \langle j_1 m_1 j_2 m_2 | j_1 m_1' \pm 1 j_2 m_2 \rangle \langle j_1 m_1' \pm 1 j_2 m_2 | J m \rangle + \sum_{m_2'} \sqrt{(j_2 \mp m_2')(j_2 \pm m_2' + 1)} \langle j_1 m_1 j_2 m_2 | j_1 m_1 j_2 m_2' \pm 1 \rangle \langle j_1 m_1 j_2 m_2' \pm 1 | J m \rangle$$

$$= \sum_{m_1'} \sqrt{(j_1 \mp m_1')(j_1 \pm m_1' + 1)} \langle j_1 m_1 j_2 m_2 | j_1 m_1' \pm 1 j_2 m_2 \rangle \langle j_1 m_1' \pm 1 j_2 m_2 | J m \rangle + \sum_{m_2'} \sqrt{(j_2 \mp m_2')(j_2 \pm m_2' + 1)} \langle j_1 m_1 j_2 m_2 | j_1 m_1 j_2 m_2' \pm 1 \rangle \langle j_1 m_1 j_2 m_2' \pm 1 | J m \rangle$$

递推：

$$J = j_1 + j_2$$

对 $m = j_1 + j_2$ ：$1 = \langle j_1 j_1 ; j_2 j_2 | j_1 + j_2, j_1 + j_2 \rangle = \langle j_1 m_1 j_2 m_2 | J m \rangle$ 此

对 $m = j_1 + j_2 - 1$

$$\langle j_1 j_1 ; j_2 (j_1+j_2-1) | J m \rangle = \sqrt{\frac{(j_1+j_2)^2 - (j_1-j_2)^2}{(j_1+j_2)^2 - (j_1+j_2-1)^2}} \langle j_1 (j_1-1) ; j_2 j_2 | J m \rangle$$

$$= \sqrt{\frac{4j_1 j_2}{2(j_1+j_2)-1}} \langle j_1 (j_1-1) ; j_2 j_2 | J m \rangle$$

以此类推，$\langle j_1 j_1 ; j_2 j_2 | J m \rangle = \sqrt{\frac{(j_1+j_2)!(j_1-j_2)!}{(2j_1)!(2j_2)!}}$ 。

3. 角动量的耦合

$$H = H_0 + H_{SO} = H_0 + \lambda \hat{S} \cdot \hat{L}$$

$\hat{H}$ 为轨态可表为轨道态与自旋态的直积 $|n l m\rangle |\sigma\rangle$

$\vec{J} = \vec{L} + \vec{S}$, $[\vec{J}, \hat{H}] = 0$

$[\hat{H}, \vec{S}^2] = 0$, $[\hat{H}, \vec{L}^2] = 0$ 这说明 $\hat{H}$ 的简并分裂为 $j = l + \frac{1}{2}$ 、 $j = l - \frac{1}{2}$ 两组.

$\hat{H} = \hat{H}_0 + \frac{\alpha}{\hbar}(\vec{J}^2 - \vec{S}^2 - \vec{L}^2)$ 简并度分别为 $2l+2$、$2l$ ($s = \frac{1}{2}, -\frac{1}{2}$)

$E = E_0 + \frac{\alpha}{2}[j(j+1) - s(s+1) - l(l+1)]$

§ 3.4 带电粒子在电磁场中的运动

$H = \frac{1}{2m}(\vec{p} - q\vec{A})^2 + q\phi$ } 正则量子化.

$\hat{H} = \frac{1}{2m}(\hat{p} - q\hat{A})^2 + q\hat{\phi}$

规范变换 $\begin{cases} \vec{A} \to \vec{A}' = \vec{A} + \nabla \chi(\vec{r}, t) \\ \phi \to \phi' = \phi - \frac{\partial}{\partial t}\chi(\vec{r}, t) \end{cases}$ $U(1)$

设 $|\psi\rangle \to |\psi'\rangle = \hat{G}|\psi\rangle$, $\langle\psi'|\psi'\rangle = \langle\psi|\psi\rangle$

$\therefore \hat{G} = e^{i\theta}$

动量算符: $\hat{\pi} = \hat{p} - q\hat{A}$

动量期待值: $\pi = \langle\psi|\hat{p} - q\hat{A}|\psi\rangle$

$= \pi' = \langle\psi'|\hat{p} - q\vec{A}|\psi'\rangle$ (为波函数不变)

$\therefore \hat{G}(\hat{p} - q\hat{A})\hat{G}^+ = \hat{p} - q\hat{A}'$ $|\psi'\rangle = \hat{G}|\psi\rangle$

$q(\hat{A}'\hat{G} - \hat{G}\hat{A}') = q\{\nabla\chi(\vec{r},t) + [\hat{A}, \hat{G}]\}$

$= \hat{p}\hat{G} - \hat{G}\hat{p} = [\hat{p}, \hat{G}]$

$[\hat{A}, \hat{G}] = 0$,

$[\hat{p}, \hat{G}]\psi = -i\hbar[\nabla(\hat{G}\psi) - \hat{G}\nabla\psi] = -i\hbar\nabla\hat{G}\cdot\psi$

$\Rightarrow q\nabla\chi(\vec{r},t) = -i\hbar\nabla\hat{G} = \hbar\nabla\theta(\vec{r},t)$,

$\therefore$ 取 $\theta(\vec{r},t) = \frac{q\chi(\vec{r},t)}{\hbar}$

$\hat{G} = e^{\pm\frac{iq\chi(\vec{r},t)}{\hbar}}$

规范变换下，schroalnger方程不变。

证：考虑规范变换 $\begin{cases} \hat{A} \to \hat{A}' = \hat{A} + \nabla \hat{\chi}(\vec{r},t) \\ \hat{\phi} \to \hat{\phi}' = \hat{\phi} - \frac{\partial}{\partial t}\hat{\chi}(\vec{r},t) \\ |\psi\rangle \to |\psi\rangle' = e^{\frac{iq\hat{\chi}(\vec{r},t)}{\hbar}}|\psi\rangle \end{cases}$

$$\left[\frac{1}{2m}(\hat{p}-q\hat{A}')^2 + q\hat{\phi}\right]|\psi'\rangle$$

$$= \frac{1}{2m}\hat{p}^2 - q\hat{p}\left[\hat{A} + \frac{\partial}{\partial t}\hat{\chi}(\vec{r},t)\right] - q\left[\hat{A} + \frac{\partial}{\partial t}\hat{\chi}(\vec{r},t)\right]\hat{p} + q^2\hat{A}^2 +$$

规范变换前后：

$$\begin{cases} \frac{1}{2m}\left[(\hat{p}-q\hat{A})^2 + q\hat{\phi}\right]|\psi\rangle = i\hbar\frac{\partial}{\partial t}|\psi\rangle \\ \frac{1}{2m}\left[\hat{p}-q\hat{A}')^2 + q\hat{\phi}'\right]|\psi'\rangle = i\hbar\frac{\partial}{\partial t}|\psi'\rangle \end{cases}$$

连续性方程：$\frac{\partial}{\partial t}\rho(\vec{r},t) + \nabla\cdot\vec{j}(\vec{r},t) = 0$，$\rho = \psi^*\psi$ 找磁流密度.

$\vec{j}(\vec{r},t) = -\frac{i\hbar}{2m}\left[\psi^*(\vec{r},t)\nabla\psi(\vec{r},t) - \psi(\vec{r},t)\nabla\psi^*(\vec{r},t)\right] - \frac{q}{m}\vec{A}\psi^*(\vec{r},t)\psi(\vec{r},t)$

设 $\psi(\vec{r},t) = \sqrt{\rho(\vec{r},t)}\, e^{\frac{i\theta(\vec{r},t)}{\hbar}}$

$\vec{j}(\vec{r},t) = \frac{\rho(\vec{r},t)}{m}\left[\nabla\theta(\vec{r},t) - q\vec{A}\right]$

Coulomb 规范 $\nabla\cdot\vec{A} = 0$ $[\hat{p},\hat{A}] = -i\hbar\nabla\cdot\vec{A} = 0$

$\hat{H} = \frac{\hat{p}^2}{2m} - \frac{q}{m}\vec{A}\cdot\hat{p} + \frac{q^2}{2m}\vec{A}^2 + q\phi$

2. Zeeman 效应.

原子处在均匀磁场 $\vec{B}$ 中.

$\hat{H} = \frac{1}{2m_e}[\hat{p} + e\vec{A}]^2 - e\phi + U(\vec{r})$

选取矢势 $\vec{A} = \frac{1}{2}\vec{B}\times\vec{r}$，$\phi = 0$

$\nabla\times\vec{A} = \frac{1}{2}\left[\partial_y(B_x y - B_y x) - \partial_z(B_z x - B_x z)\cdots\right]$

$= \frac{1}{2}(2B_x, 2B_y, 2B_z) = \vec{B}$

$\nabla\cdot\vec{A} = \frac{1}{2}(\nabla\cdot\vec{B})\times\vec{r} = 0$

$\hat{H} = \hat{H}_0 + \frac{e}{m_e}\hat{p}\vec{A} + \frac{e^2}{2m_e}\vec{A}^2$

$= \hat{H}_0 + \frac{e}{2m_e}\hat{p}\cdot(\vec{B}\times\vec{r}) + \frac{e^2}{8m_e}(\vec{B}\times\vec{r})^2$ $\hat{p}\cdot(\vec{B}\times\vec{r}) = \vec{B}\cdot(\vec{r}\times\hat{p}) = \vec{B}\cdot\hat{L}$

$= \hat{H}_0 + \frac{e}{2m_e}\vec{B}\cdot\hat{L} + \frac{e^2}{8m_e}(\vec{B}\times\vec{r})^2$

$\approx \hat{H}_0 + \frac{e}{2m_e}\vec{B}\cdot\hat{L} = \hat{H}_0 - \hat{\mu}_L\cdot\vec{B}$ Bohr 磁子 $\hat{\mu}_L = -\frac{e\hat{L}}{2m_e}$

$\frac{\text{第二项}}{\text{第一项}} \sim \frac{eBa^2}{\hbar} \ll 1$

角动与磁场相互作用：$\hat{H}_S = -\hat{\mu}_S\cdot\vec{B}$ $\hat{\mu}_S = -\frac{e}{m_e}\hat{S}$

g 因子：$\begin{cases}\hat{\mu}_L = -g_L\frac{\mu_B}{\hbar}\hat{L}, & g_L = 2\\ \hat{\mu}_S = -g_S\frac{\mu_B}{\hbar}\vec{S}, & g_S = 1\end{cases}$

3. Landau 能级

(1) Landau 规范 具有平移不变性.

粒子在沿 z 方向的均匀磁场 $\vec{B} = B\vec{e}_z$ 中运动

选取规范：$\vec{A} = (0, Bx, 0)$

$\hat{H} = \frac{1}{2m}\hat{p}_x^2 + \frac{1}{2m}(\hat{p}_y - qB\hat{x})^2 + \frac{1}{2m}\hat{p}_z^2$

$[\hat{p}_y, \hat{H}] = 0$, $[\hat{p}_z, \hat{H}] = 0$ → 说明 y, z 方向为平移对称性.

$\psi(x,y,z) = \phi(x)\, e^{\frac{i}{\hbar}(p_y y + p_z z)}$

Schrodinger 方程: $\hat{H}\psi = E\psi$

$$\frac{1}{2m}\left[\hat{p}_x^2 + (\hat{p}_y - qBx)^2 + \hat{p}_z^2\right]\psi = E\psi$$

$$= \frac{1}{2m}\left(-\hbar^2 \frac{\partial^2}{\partial x^2}\psi\right) + \frac{1}{2m}\left[(p_y - qBx)^2 + p_z^2\right]\psi = E\psi$$

$$= -\frac{\hbar^2}{2m}\frac{d^2\phi(x)}{dx^2} + \frac{q^2B^2}{2m}\left(x - \frac{p_y}{qB}\right)^2\phi(x) = \left(E - \frac{p_z^2}{2m}\right)\phi(x) = E\phi(x)$$

$\omega = \frac{qB}{m}$, $x_0 = \frac{p_y}{qB}$

$$E = -\frac{\hbar^2}{2m}\frac{d^2\phi(x)}{dx^2} + \frac{1}{2}m\omega^2(x - x_0)^2\phi(x) = \left(E - \frac{p_z^2}{2m}\right)\phi(x)$$

⇒ $E_n = \left(n + \frac{1}{2}\right)\hbar\omega + \frac{p_z^2}{2m} = \left(n + \frac{1}{2}\right)\frac{qB\hbar}{m} + \frac{p_z^2}{2m}$, 为 Landau 能级.

相应本征函数:

$$\psi_n(x,y,z) = \frac{1}{2^n n!}\left(\frac{\beta}{\pi}\right)^{1/2} H_n[\beta(x - x_0)] \exp\left[-\frac{\beta^2}{2}(x - x_0)^2\right] e^{\frac{i}{\hbar}(p_y y + p_z z)}$$

简并度: $D = 2 \cdot \frac{\Phi}{\Phi_0}$, $\Phi_0 = \frac{2\pi\hbar}{e}$, 为磁通量子

a) 对称规范

考察均匀磁场中进行二维运动的粒子

$$\hat{H} = \frac{1}{2m}\left[(\hat{p}_x - qA_x)^2 + (\hat{p}_y - qA_y)^2\right] = \frac{1}{2m}\left(\hat{\pi}_x^2 + \hat{\pi}_y^2\right)$$

$$[\hat{\pi}_x, \hat{\pi}_y] = [\hat{p}_x - qA_x, \hat{p}_y - qA_y]$$

$$= q\left(\hat{p}_y A_x - \hat{p}_x A_y\right) = i\hbar q\left(\frac{\partial}{\partial x}A_y - \frac{\partial}{\partial y}A_x\right)$$

$$= i\hbar q B_z$$

定义产生算符 $\hat{a} = \frac{1}{\sqrt{2\hbar qB}}(\hat{\pi}_x + i\hat{\pi}_y)$,

湮灭算符 $\hat{a}^+ = \frac{1}{\sqrt{2\hbar qB}}(\hat{\pi}_x - i\hat{\pi}_y)$

⇒ $\hat{\pi}_x = \sqrt{\frac{\hbar qB}{2}}(\hat{a} + \hat{a}^+)$, $\hat{\pi}_y = -i\sqrt{\frac{\hbar qB}{2}}(\hat{a} - \hat{a}^+)$

$$\hat{a}^+\hat{a} = \frac{1}{2\hbar qB}\left[\hat{\pi}_x^2 + \hat{\pi}_y^2 + i(\hat{\pi}_x\hat{\pi}_y - \hat{\pi}_y\hat{\pi}_x)\right]$$

$$= \frac{1}{2\hbar qB}\left[\hat{\pi}_x^2 + \hat{\pi}_y^2 - \hbar qB\right) = \frac{m}{\hbar qB}\cdot\frac{1}{2m}\left(\hat{\pi}_x^2 + \hat{\pi}_y^2\right) - \frac{1}{2}$$

$$\hat{H} = \hbar\omega\left(\hat{a}^+\hat{a} + \frac{1}{2}\right) \qquad \omega = \frac{qB}{m}$$

$E_n = \hbar\omega\left(n + \frac{1}{2}\right)$, Landau 能级.

对之有化为：

$$\vec{A} = \frac{B}{2}(-\vec{y}, \vec{x}, 0)$$

$$\begin{aligned}\hat{H} &= \frac{1}{2m}\left\{(\hat{p}_x - q\hat{A}_x)^2 + (\hat{p}_y - q\hat{A}_y)^2\right\} \\ &= \frac{1}{2m}\left\{(\hat{p}_x + \frac{1}{2}qB\hat{y})^2 + (\hat{p}_y - \frac{1}{2}qB\hat{x})^2\right\} \\ &= \frac{1}{2m}\left(\hat{p}_x^2 + qB\hat{p}_x\hat{y} + \frac{1}{4}q^2B^2\hat{y}^2 + \hat{p}_y^2 - qB\hat{p}_y\hat{x} + \frac{1}{4}q^2B^2\hat{x}^2\right) \\ &= \frac{1}{2m}\left(\hat{p}_x^2 + \hat{p}_y^2\right) - \frac{qB}{2m}\hat{L}_z + \frac{q^2B^2}{8m}(\hat{x}^2 + \hat{y}^2)\end{aligned}$$

有关之轴的旋转对称性。

$[\hat{L}_z, \hat{H}] = 0 \Rightarrow$ 角动量本征态可表为 $|n, m\rangle$

$$\hat{H}|n,m\rangle = E_{n,m}|n,m\rangle, \quad \hat{L}_z|n,m\rangle = m\hbar|n,m\rangle$$

定义 $\vec{\lambda}' = \hat{p} + q\vec{A}$

(1) $[\hat{\lambda}_x', \hat{\lambda}_y'] = -i\hbar qB$

(2) $[\lambda_x, \lambda_x'] = [\lambda_y, \lambda_y'] = [\lambda_x, \lambda_y'] = [\lambda_y, \lambda_x'] = 0$

(3) $\hat{L}_z = \frac{1}{2qB}(\hat{\lambda}_x'^2 + \hat{\lambda}_y'^2 - \hat{\lambda}_x^2 - \hat{\lambda}_y^2)$

$\hat{\lambda}_x'^2 - \hat{\lambda}_x^2 \to \hat{\lambda}_y'^2 - \hat{\lambda}_y^2$

$= 4q(\hat{p}_x\hat{A}_x + \hat{p}_y\hat{A}_y) = 2qB[\hat{p}_x(-\hat{y}) + \hat{p}_y\hat{x}]$

定义 $\hat{b} = \frac{1}{\sqrt{2\hbar qB}}(\hat{\lambda}_x' + i\hat{\lambda}_y')$, $\hat{b}^+ = \frac{1}{\sqrt{2\hbar qB}}(\hat{\lambda}_x' - i\hat{\lambda}_y')$

$\hat{\lambda}_x' = \sqrt{\frac{\hbar qB}{2}}(\hat{b} + \hat{b}^+)$, $\hat{\lambda}_y' = -\sqrt{\frac{\hbar qB}{2}}i(\hat{b} - \hat{b}^+)$

$[\hat{b}, \hat{b}^+] = \frac{1}{2\hbar qB}(-2i)[\lambda_x', \lambda_y'] = 1$

$[\hat{a}, \hat{b}] = \frac{1}{2\hbar qB}\{[\lambda_x, \lambda_x'] + i([\lambda_y, \lambda_x'] + [\lambda_x', \lambda_y]) - [\lambda_y, \lambda_y']\} = 0$

$[\hat{a}^+, \hat{b}^+] = 0$

$$\begin{aligned}\hat{L}_z &= \frac{\hbar}{4}\left\{(\hat{b} + \hat{b}^+)^2 - (\hat{b} - \hat{b}^+)^2 - (\hat{a} + \hat{a}^+)^2 + (\hat{a} - \hat{a}^+)^2\right\} \\ &= \hbar(\hat{b}^+\hat{b} - \hat{a}^+\hat{a})\end{aligned}$$

$\hat{L}_z(\hat{b}^+|n,m\rangle) = (m+1)\hbar|n,m\rangle$, $\hat{L}_z(\hat{a}^+|n,m\rangle) = (n-1)\hbar|n,m\rangle$

$$\Rightarrow |n,m\rangle = \frac{\hat{a}^+}{\sqrt{n!}}\frac{(\hat{b}^+)^{n+m}}{\sqrt{(n+m)!}}|0,0\rangle$$

求考虑态的具体形式，$\omega = x + iy$

$$\psi = \sqrt{\frac{qB}{\hbar}}\frac{1}{\sqrt{2^{n+m}n!(n+m)!}}\omega^m e^{-\frac{qB}{4\hbar}|\omega|^2}\left(\frac{qB}{2\hbar}\omega - 2\frac{\partial}{\partial\omega}\right)^{n+m}1$$

$$\begin{cases} \hat{a} = \sqrt{\frac{\hbar}{2m\omega}}\left(2\frac{\partial}{\partial w^*} + \frac{m\omega}{\hbar} w\right) \\ \hat{a}^\dagger = -\sqrt{\frac{\hbar}{2m\omega}}\left(2\frac{\partial}{\partial w} - \frac{m\omega}{\hbar} w^*\right) \end{cases}$$

$$\begin{cases} \hat{b} = \sqrt{\frac{\hbar}{2m\omega}}\left(2\frac{\partial}{\partial w} + \frac{m\omega}{\hbar} w\right) \\ \hat{b}^\dagger = \sqrt{\frac{\hbar}{2m\omega}}\left(-2\frac{\partial}{\partial w^*} + \frac{m\omega}{\hbar} w^*\right) \end{cases}$$

$$= \frac{1}{\sqrt{2m\hbar\omega}}\left[\hat{p}_x + i\hat{p}_y + m\omega(\hat{x} + i\hat{y})\right]$$

$$= -i\sqrt{\frac{\hbar}{2m\omega}}\left[\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right) + \frac{m\omega}{\hbar}(x + iy)\right]$$

$$= -i\sqrt{\frac{\hbar}{2m\omega}}\left(2\frac{\partial}{\partial w^*} + \frac{m\omega}{\hbar} w\right)$$

基态 $|0,0\rangle$ 满足

$$\hat{a}|0,0\rangle = 0, \quad \hat{b}|0,0\rangle = 0$$

$$\langle w, w^*|\hat{a}|0,0\rangle = -i\sqrt{\frac{\hbar}{2m\omega}}\left(2\frac{\partial}{\partial w^*} + \frac{m\omega}{\hbar} w\right)\langle w, w^*|0,0\rangle = 0$$

解得 $\psi_{0,0}(w, w^*) = \langle w, w^*|0,0\rangle = f(w)\, e^{-\frac{m\omega}{4\hbar}|w|^2}$

$$\langle w, w^*|\hat{b}|0,0\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(2\frac{\partial}{\partial w} + \frac{m\omega}{\hbar} w^*\right)\langle w, w^*|0,0\rangle = 0$$

解得 $f(w) = C$

$$\Rightarrow \psi_{0,0}(w, w^*) = C_{0,0}\, e^{-\frac{m\omega}{4\hbar}|w|^2}$$

$$\psi_{n,m}(w, w^*) = \langle w, w^*|n, m\rangle$$

$$= \frac{1}{\sqrt{n!(n+m)!}}\left[-i\sqrt{\frac{\hbar}{2m\omega}}\left(2\frac{\partial}{\partial w} - \frac{m\omega}{\hbar} w\right)\right]^n \left[\sqrt{\frac{\hbar}{2m\omega}}\left(-2\frac{\partial}{\partial w^*} + \frac{m\omega}{\hbar} w^*\right)\right]^{n+m} \psi_{0,0}(w, w^*)$$

$$= C_{n,m}\left(\frac{\partial}{\partial w} - \frac{m\omega}{4\hbar} w^*\right)^n \left(w^{n+m}\, e^{-\frac{m\omega}{4\hbar}|w|^2}\right)$$

AB效应

引入磁场前后，具有相同能量的本征函数 $\phi(\vec{r})$，$\psi(\vec{r})$

$$\psi(\vec{r}) = \phi(\vec{r})\, e^{\frac{iq}{\hbar}\int_0^{\vec{r}} A(\vec{r}')\,d\vec{r}'}$$

证：已知 $\left[\frac{1}{2m}(\vec{p}-q\vec{A})^2 + V(\vec{r})\right]\psi(\vec{r}) = E\psi(\vec{r})$

$$= \left[\frac{1}{2m}(\vec{p}-q\vec{A})^2 + V(\vec{r})\right]\phi(\vec{r})\, e^{\frac{iq}{\hbar}\int_0^{\vec{r}} A(\vec{r}')\,d\vec{r}'} \quad \{\vec{p}\cdot\phi(\vec{r})\}$$

$$= e^{\frac{iq}{\hbar}\int_0^{\vec{r}} A(\vec{r}')\,d\vec{r}'}\left[\frac{1}{2m}(\vec{p}-q\vec{A}+q\vec{A})^2 + V(\vec{r})\right]\phi(\vec{r})$$

$$= E\phi(\vec{r})$$

$$\psi(\vec{r}) = \psi^{(0)}(\vec{r}) + \psi^{(1)}(\vec{r})$$

$$= e^{\frac{iq}{\hbar}\int_{\vec{r}_0}^{\vec{r}} A\cdot d\vec{l}}\left(\phi_0^{(0)} + \phi_0^{(1)} e^{\frac{iq}{\hbar}\oint A\,d\vec{l}}\right), \quad \oint A\cdot d\vec{l} = \int_s \vec{B}\cdot\vec{n}\,ds = \Phi$$

重点 - 介规范不变量.

磁通量子化：

相位差 $e^{\frac{iq}{\hbar}\Phi}$ 导出 $\frac{q\Phi}{\hbar} = 2n\pi$

$\Rightarrow \Phi_0 = \frac{2\pi\hbar}{q}$

解法：

1. $r > a$ ，$\psi = 0$

$r < \frac{a}{2}$ ，$\frac{d^2u}{dr^2} + k^2 u(r) = 0$ ，$k = \sqrt{\frac{2mE}{\hbar}}$ ，$u_1(r) = A\sin kr + B\cos kr$

$\lim_{r\to 0} r^2 U(r) = 0 \Rightarrow \cos r$ 不满足，$u_1(r) = A\sin kr$

$\frac{a}{2} < r < a$ ，$\frac{d^2u}{dr^2} - \kappa^2 r = 0$ ，$\kappa = \sqrt{\frac{2m(U_0 - E)}{\hbar}}$

$u_2(r) = Be^{\kappa r} + Ce^{-\kappa r}$

由
$$\begin{cases} A\sin\left(k\frac{a}{2}\right) = Be^{\kappa\frac{a}{2}} + Ce^{-\kappa\frac{a}{2}} \\ kA\cos\left(k\frac{a}{2}\right) = \kappa Be^{\kappa\frac{a}{2}} - \kappa Ce^{-\kappa\frac{a}{2}} \\ Be^{\kappa a} + Ce^{-\kappa a} = 0 \end{cases}$$

$C = -Be^{2\kappa a}$

$$\begin{cases} A\sin\left(\frac{ka}{2}\right) - Be^{\kappa\frac{a}{2}}\left(1 - e^{\kappa a}\right) = 0 \\ kA\cos\left(\frac{ka}{2}\right) - B\kappa e^{\kappa\frac{a}{2}}\left(1 + e^{\kappa a}\right) = 0 \end{cases}$$

非平凡解：$\begin{vmatrix} \sin\frac{ka}{2} & -e^{\kappa\frac{a}{2}}\left(1 - e^{\kappa a}\right) \\ k\cos\frac{ka}{2} & -\kappa e^{\kappa\frac{a}{2}}\left(1 + e^{\kappa a}\right) \end{vmatrix} = \kappa\sin\frac{ka}{2}\left(1 + e^{\kappa a}\right) - k\cos\frac{ka}{2}\left(1 - e^{\kappa a}\right) = 0$

$$\frac{\tan\frac{ka}{2}}{k} = -\frac{\tanh\frac{\kappa a}{2}}{\kappa}$$

$$k = -\kappa\frac{\tan\frac{ka}{2}}{\tanh\frac{\kappa a}{2}} > 0$$

$\frac{ka}{2} \in \left(\frac{2n-1}{2}\pi, n\pi\right)$

简约 $k \in \left(\frac{2n-1}{a}\pi, \frac{2n}{a}\pi\right)$ ，$E \in \left(\frac{(2n-1)^2\pi^2\hbar^2}{2ma^2}, \frac{n^2\pi^2\hbar^2}{2ma^2}\right)$

$n = 1, 2 \cdots n_{max}$, $n_{max} = \left\lfloor \frac{m_e v_e}{\hbar} \right\rfloor$

2. (1)

(2) $\psi = R Y$

$R = C e^{-\frac{r}{a}}$

径方向密度 $\rho(r) = R^2 r^2$

$\frac{d\rho(r)}{dr} = 0 \rightarrow r = a$

(3) $R^2(r)$

3、 时间演化算符 $\hat{U} = \begin{pmatrix} e^{-i\frac{\varepsilon_1}{\hbar}t} & 0 \\ 0 & e^{-i\frac{\varepsilon_2}{\hbar}t} \end{pmatrix}$

$$\begin{cases} -\frac{\varepsilon_1}{\hbar}t = 2k_1\pi \\ -\frac{\varepsilon_2}{\hbar}t = 2k_2\pi + \varphi \end{cases}$$

$$H = \begin{pmatrix} \varepsilon_1 & 0 \\ 0 & \varepsilon_2 \end{pmatrix} = \frac{\hbar\omega}{2}\left(N + \frac{\varphi}{2\pi}\right)\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + C(\omega, \varphi)$$

(2) $\hat{Q}^+ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$$\hat{Q}^+ \hat{U}_{NOT} \hat{Q} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\begin{cases} \varepsilon_1 \frac{\Delta t}{\hbar} = 2k\pi \\ \varepsilon_2 \frac{\Delta t}{\hbar} = (2m+1)\pi \end{cases}$$

$$H' = \frac{\hbar\omega}{2}\left(N + \frac{1}{2}\right)\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\hat{H} = \hat{Q} H' \hat{Q}^+ = \frac{\hbar\omega}{2}(N+1)\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

4、(1) 选取基矢：$|\uparrow\uparrow\rangle$，$|\uparrow\downarrow\rangle$，$|\downarrow\uparrow\rangle$，$|\downarrow\downarrow\rangle$（直接）

$$\hat{H} = J\hat{S}_1\hat{S}_2 + [J_0 - J]\hat{S}_2^z$$

$$\{\hat{S}_1^z \hat{S}_2^z, \hat{S}_1^x \hat{S}_2^x\}$$

$$= \hat{S}_1^z \hat{S}_2^z \hat{S}_1^x \hat{S}_2^x - \hat{S}_1^x \hat{S}_2^x \hat{S}_1^z \hat{S}_2^z$$

不同角度之间为互换

$$= \hat{S}_1^z \hat{S}_1^x \hat{S}_2^z \hat{S}_2^x - \hat{S}_1^x \hat{S}_1^z \hat{S}_2^x \hat{S}_2^z$$

反对易.

$$=$$

$= 0$

$[\hat{S}_1^{(1)}\hat{S}_2^{(1)}, \hat{S}_1^{(2)}\hat{S}_2^{(2)}]$

∴ $\hat{H}$ 为用 $\hat{S}_1\hat{S}_2$, $\hat{S}_1^{(2)}\hat{S}_2^{(2)}$ 的共同本征态表示.

5. (1) 归一化 $A = \frac{1}{\sqrt{6}}$

$\langle \hat{L}_x \rangle = \frac{1}{6}\times 0 + \frac{1}{6}\hbar + \frac{2}{3}(-\hbar) = -\frac{1}{2}\hbar$

$\langle \hat{S}_x \rangle = -\frac{1}{6}\hbar$

(2) $[\hat{L}_x, \hat{L}^2] = 0$ → 可按 $\hat{L}_x$ 共同本征态分类.

① $l=0$ $|100\rangle$ 为 $\hat{L}_x$ 本征态, 本征值为 0.

$\langle 100, \uparrow | \psi \rangle = \frac{1}{\sqrt{6}}$, $\langle 100, \downarrow | \psi \rangle = 0$

② $l=1$ 选取 $\{|121\rangle, |120\rangle, |121\rangle\}$ 为基矢

$\hat{L}_x = \frac{\sqrt{2}}{2}\hbar \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ → $L_x = \pm\hbar, 0$

$\langle 2,1, L_x=-\hbar, \uparrow | \psi \rangle = \frac{1}{2\sqrt{6}}$, $\langle 2,1, L_x=-\hbar, \downarrow | \psi \rangle = \frac{1}{\sqrt{6}}$

$\langle 2,1, L_x=0, \uparrow | \psi \rangle = \frac{1}{2\sqrt{3}}$, $\langle 2,1, L_x=0, \downarrow | \psi \rangle = \frac{1}{\sqrt{3}}$

$\langle 2,1, L_x=\hbar, \uparrow | \psi \rangle = \frac{1}{2\sqrt{6}}$, $\langle 2,1, L_x=\hbar, \downarrow | \psi \rangle = \frac{1}{\sqrt{6}}$

$P(L_x=-\hbar) = \frac{5}{24}$

$P(L_x=0) = \frac{1}{12}$

$P(L_x=\hbar) = \frac{5}{24}$

(3) $\langle \hat{L}_z^2 \rangle = \hbar^2$ → $\langle \hat{L}_z \rangle = \pm\hbar$

$\begin{pmatrix} \psi_{11} \\ \end{pmatrix}$

$\frac{1}{\sqrt{2}}$ $2\psi_{n-1}$

b. 选取规范 $\vec{A} = (0, Bx, 0)$

$\hat{p}_y' = \hat{p}_y + eB\hat{x}$

$\hat{H}' = \begin{pmatrix} 0 & \hat{p}_x - i\hat{p}_y - 2eB\hat{x} \\ \hat{p}_x + i\hat{p}_y + ieB\hat{x} & 0 \end{pmatrix}$

本征态 $\hat{H}|\psi\rangle = E|\psi\rangle$, $\hat{H}\begin{pmatrix} \psi_x(x,y) \\ \psi_y(x,y) \end{pmatrix} = E\begin{pmatrix} \psi_x(x,y) \\ \psi_y(x,y) \end{pmatrix}$

{

$\Rightarrow v_f^2\left[\hat{p}_x^2 + \hat{p}_y^2 - eB\hbar + 2eB\hat{x}\hat{p}_y + e^2B^2\hat{x}^2\right]\psi_y(x,y) = E^2\psi_y(x,y)$

平移对称性: $\psi_y(x,y) = \mu(x)e^{-\frac{i}{\hbar}p_y y}$

$\left[v_f^2\hat{p}_x^2 + v_f^2 e^2B^2\left(x + \frac{p_y}{eB}\right)^2\right]\mu(x) = (E^2 + eB\hbar v_f^2)\mu(x)$

$m = \frac{1}{2v_f^2}$, $\omega = 2v_f^2 eB\hbar$

$E^2 + eB\hbar v_f^2 = (n+\frac{1}{2})\omega$

$E = \pm v_f\sqrt{2eB\hbar n}$

$\hat{\pi}_x = \hat{p}_x + e\hat{A}_x$, $\hat{\pi}_y = \hat{p}_y + e\hat{A}_y$

{ $\hat{a} = \frac{1}{\sqrt{2\hbar eB}}(\hat{\pi}_x + i\hat{\pi}_y)$

$$\vec{a}^{\dagger} = \frac{1}{\sqrt{2\hbar eB}}(\vec{\pi}_x - i\vec{\pi}_y)$$

$$\vec{H} = v_F\sqrt{2\hbar eB}\begin{pmatrix} 0 & \vec{a} \\ \vec{a}^{\dagger} & 0 \end{pmatrix} = \hbar\omega\begin{pmatrix} 0 & \vec{a} \\ \vec{a}^{\dagger} & 0 \end{pmatrix}$$

$$\vec{H}|\psi\rangle = z|\psi\rangle$$

$$\vec{H}^2|\psi\rangle = z^2|\psi\rangle \qquad \vec{H}^2 = \hbar^2\omega^2\begin{pmatrix} \vec{a}\vec{a}^{\dagger} & 0 \\ 0 & \vec{a}^{\dagger}\vec{a} \end{pmatrix}$$

$$\begin{cases} \vec{H}^2|\psi_1\rangle = \hbar^2\omega^2\,\vec{a}\vec{a}^{\dagger}|\psi_1\rangle = \hbar^2\omega^2(\vec{a}^{\dagger}\vec{a}+1)|\psi_1\rangle = z^2|\psi_1\rangle \\ \vec{H}^2|\psi_2\rangle = \hbar^2\omega^2\,\vec{a}^{\dagger}\vec{a}|\psi_2\rangle = z^2|\psi_2\rangle \end{cases}$$

$$\Rightarrow \begin{cases} |\psi_1\rangle = |n-1\rangle \\ |\psi_2\rangle = |n\rangle \end{cases}, \quad z_n = \pm\sqrt{n}\,\hbar\omega, \quad |\psi_n\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} |n-1\rangle \\ |n\rangle \end{pmatrix}$$